

MBS-009 - Relation to AI and ASI

How MET Brain Structure Emerged from Structural Intelligence Research

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Repository: <https://github.com/sizhet/MET-Brain-Structure-MBS>

Introduction

At first glance, the MET Brain Structure (MBS) project may appear to be a neuroscience-oriented effort.

However, its origins are surprisingly different.

MBS did not emerge from neuroscience research alone.

It emerged from a long sequence of investigations into:

- Artificial Intelligence
- Structural Intelligence
- Autonomous Systems
- Calling Graph architectures
- Function Tunnel Intelligence
- Runtime adaptation
- Human-AI collaboration

In many ways, MBS is the result of an unexpected journey.

Researchers were attempting to understand intelligent systems.

Along the way, they repeatedly encountered structural patterns that eventually appeared relevant to biological cognition itself.

This document explains that journey.

The Original Question

The original question was not:

How does the brain work?

Instead, the original question was:

How can intelligent systems organize and evolve complex behavior?

This question led to investigations involving:

- Search structures

- Knowledge organization
- State transitions
- Action pathways
- Runtime adaptation
- Structural evolution

Over time, a collection of ideas emerged under the broader umbrella of Structural Intelligence.

Structural Intelligence

Structural Intelligence proposes that intelligence may not be reducible to isolated symbols or numerical optimization alone.

Instead, intelligence may depend heavily upon:

- Structure
- Relationships
- Paths
- Constraints
- Evolutionary continuity

This perspective led to the development of several frameworks.

Examples include:

- Differential Trees
- Common Concept Core (CCC)
- Calling Graph systems
- Function Tunnel Intelligence (FTI)
- Autonomous Structural Intelligence (ASI)

Initially these frameworks were intended primarily for computational and engineering applications.

However, their implications gradually expanded.

Calling Graphs

One of the most influential concepts was the Calling Graph.

A Calling Graph describes transitions among states through operations.

State

→ Operation

→ State

At first this appeared to be a software architecture concept.

Yet over time an intriguing observation emerged.

Many biological behaviors also appear to involve:

- States
- Transitions
- Action sequences

- Functional continuity

This observation did not prove any connection.

But it suggested that graph-based structures might offer useful cognitive abstractions.

Function Tunnel Intelligence

Function Tunnel Intelligence introduced another important idea.

Rather than viewing behavior as isolated actions, FTI emphasized:

Functional pathways.

Behavioral continuity.

Trajectory preservation.

Tunnel-constrained evolution.

The central insight was that many intelligent systems appear to evolve through stable functional pathways rather than arbitrary transitions.

Once again, parallels began to appear.

Animals.

Humans.

Organizations.

Civilizations.

Many systems seemed to rely on persistent functional trajectories.

This raised a new possibility:

Perhaps biological cognition also relies on extensive networks of functional tunnels.

Autonomous Structural Intelligence

The development of Autonomous Structural Intelligence (ASI) pushed these ideas further.

ASI focused on systems capable of:

- Structural adaptation
- Runtime learning
- Path selection
- Autonomous evolution

A recurring challenge emerged.

For adaptation to occur, a system must continuously answer:

What matters?

What should be activated?

What should be ignored?

These questions repeatedly led back to:

Triggering.

Retrieval.

Structural activation.

The same themes appeared again and again.

The Unexpected Return to Biology

At this point, an unexpected pattern became difficult to ignore.

Several observations seemed strangely compatible:

Dreaming.

Threat response.

Memory recall.

Navigation.

Skill acquisition.

Imagination.

Planning.

Each appeared to involve:

Activation.

Retrieval.

Pathways.

Scenes.

Adaptation.

The similarities were not proof.

However, they were sufficient to motivate a structural hypothesis.

The result eventually became the MET Brain Structure project.

MBS as a Reverse Discovery

Many theories begin with biology and move toward computation.

MBS followed the opposite path.

The progression was approximately:

Engineering Structures

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Runtime Systems

↓

Calling Graphs

↓

Function Tunnels

↓

Autonomous Adaptation

↓

Cognitive Analogies

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Brain Structure Hypothesis

In this sense, MBS can be viewed as a reverse discovery. Rather than designing AI from neuroscience, it explores whether certain engineering structures unexpectedly resemble aspects of biological cognition.

Relationship to Large Language Models

Large Language Models have demonstrated extraordinary capabilities.

Language understanding.

Generation.

Reasoning.

Coding.

Knowledge retrieval.

However, LLMs also reveal limitations.

They are primarily language-centered systems.

Many aspects of biological cognition appear broader.

For example:

- Embodied experience
- Scene construction
- Persistent goals
- Continuous adaptation
- Trigger-driven behavior
- Physical interaction

The MBS perspective therefore suggests that future intelligence systems may require additional architectural components beyond language models alone.

This is not a criticism of LLMs.

Rather, it is a recognition that language may represent only one layer of cognition.

Trigger Systems and AI

Most contemporary AI systems place heavy emphasis on representation and inference.

MBS highlights another possibility:

Triggering may be equally important.

An intelligent system must decide:

What deserves activation?

What deserves retrieval?

What deserves attention?

Future AI architectures may require increasingly sophisticated trigger systems capable of managing large structural knowledge spaces.

Embodied Scene Runtime and AI

Current AI systems often operate through symbolic or token-based representations.

MBS suggests that future systems may increasingly rely on:

- Scene activation
- Context reconstruction
- Runtime simulation
- Functional pathways

The Embodied Scene Runtime concept therefore represents a potential direction for future AI development.

Scenes become operational structures rather than passive representations.

Function Tunnels and Autonomous Agents

Autonomous agents require:

- Goal maintenance
- Behavioral continuity
- Adaptation
- Recovery from failure

These requirements closely resemble Function Tunnel concepts.

Future agent architectures may benefit from explicit representations of:

- Functional pathways
- Tunnel activation
- Tunnel refinement
- Tunnel evolution

The relationship between MBS and FTI becomes particularly strong at this point.

Relationship to Autonomous Structural Intelligence

Among all Structural Intelligence projects, ASI may be the closest relative of MBS.

Both focus on:

- Runtime behavior
- Structural adaptation
- Continuous evolution
- Triggering
- Retrieval
- Path activation

The primary difference lies in domain.

ASI focuses on artificial systems.

MBS focuses on biological cognition.

The underlying structural questions, however, are often remarkably similar.

Human-AI Hybrid Systems

The Human-AI Hybrid Civilization perspective introduces yet another connection.

Future intelligence may not belong exclusively to humans or machines.

Instead, intelligence may emerge from interactions among:

- Humans
- AI systems
- Shared knowledge structures
- Collective triggering mechanisms
- Distributed functional tunnels

Understanding biological cognition may therefore help guide the design of future hybrid intelligence systems.

MBS as a Bridge

The most useful way to view MBS may be as a bridge.

One side of the bridge contains:

Neuroscience.

Cognitive science.

Behavioral research.

The other side contains:

AI.

Structural Intelligence.

Autonomous Systems.

Runtime architectures.

MBS attempts to create a common structural language through which ideas can travel in both directions.

Looking Forward

The MBS project remains speculative.

Many aspects require validation.

Many assumptions may prove incomplete.

However, its value may not depend entirely upon correctness.

Even imperfect structural models can reveal useful questions.

They can expose hidden similarities.

They can encourage interdisciplinary collaboration.

And they can inspire new computational experiments.

This is the primary goal of MBS.

Conclusion

The MET Brain Structure project emerged not from an attempt to replace neuroscience, but from a long effort to understand intelligent systems.

Calling Graphs.

Function Tunnel Intelligence.

Autonomous Structural Intelligence.

Runtime adaptation.

Structural evolution.

These investigations gradually revealed patterns that appeared increasingly relevant to biological cognition.

MBS represents an attempt to explore those patterns.

It stands at the intersection of neuroscience and artificial intelligence.

It learns from both.

It belongs exclusively to neither.

Its purpose is simple:

To investigate whether a common structural language might help us better understand intelligence in all of its forms.